

Solution Midterm Exam Fall 2022

Q1. Write programs according to given description:

(10 x 6 = 60 Points)

- a. Write a program to input value of x. Further write code to evaluate and print the expression: $\frac{3x^3 - 5x^2}{2x}$

```
printf ("X: ");
scanf ("%d", &x);
r = (3 * x * x * x - 5 * x * x) / (2 * x);
printf ("Ans: %d\n", r);
```

- b. Write a program to input two numbers. If first number is smaller than second number, print integers from first number to second number, excluding second number. If first number is equal or larger then give appropriate message(s). See sample run for your understanding:

```
printf ("N1: ");
scanf ("%d", &n1);
printf ("N2: ");
scanf ("%d", &n2);
if (n1 == n2)
    printf ("N1 == N2\n");
else if (n1 > n2)
    printf ("N1 > N2");
while (n1 < n2){
    printf ("%d ", n1);
    n1 = n1 + 1;
}
printf("\n");
```

Sample Run:

```
N1: 20
N2: 20
N1 = N2
N1: 20
N2: 18
N1 > N2
N1: 20
N2: 24
20 21 22 23
```

- c. Write a program-using while loop. A student has appeared in the exams. There are total 10 subjects. He has failed to appear in some subjects. Input his marks in the subject, only if he has appeared in the exam and find average marks in 10 subjects and average marks in the subjects, he has appeared in the exam. See sample run for your understanding:

```
count = sum = 0;
while (i <= 10){
    printf ("Exam {%d} : ", i);
    scanf (" %c", &exam);
    if (exam == 'Y' || exam == 'y'){
        printf("Marks: ");
        scanf("%d", &marks);
        sum = sum + marks;
        count = count + 1;
    }
    i = i + 1;
}
printf ("Average Marks: %d\n", sum / 10);
printf ("Average Marks in appeared exams: %d\n", sum / count);
```

Sample Run:

```
Exam 1: Y
Marks: 85
Exam 2: Y
Marks: 75
Exam 3: N
Exam 4: Y
...
Average Marks: 44
Average Marks in
appeared exams: 65
```

- d. A company delivers snacks in packs only. The pack size is eight snacks per pack. They received order of snacks in bulk. However, they calculate the number of packs required. If the last pack has snacks less than eight snacks, they deliver a full pack without extra charges of extra snacks. You have to write program for input three order of snacks and calculate extra snacks in each order and total extra snacks in three orders. See sample run for your understanding:

```
printf("Order 1: ");
scanf ("%d", &order1);
printf("Order 2: ");
scanf ("%d", &order2);
printf("Order 3: ");
scanf ("%d", &order3);
extra1 = ((order1 - 1) / 8 + 1) * 8 - order1;
extra2 = ((order2 - 1) / 8 + 1) * 8 - order2;
extra3 = ((order3 - 1) / 8 + 1) * 8 - order3;
```

Sample Run:

```
Order 1: 85
Order 2: 122
Order 3: 166
Extra Snacks 1: 3
Extra Snacks 2: 6
Extra Snacks 3: 2
Total Extra Snacks: 11
```

```
printf ("Extra Snacks 1: %d\n", extra1);
printf ("Extra Snacks 2: %d\n", extra2);
printf ("Extra Snacks 3: %d\n", extra3);
printf ("Total Extra Snacks: %d\n", extra1 + extra2 + extra3);
```

- e. Write code to help your friend in packing. He has three rectangular boxes of different sizes. You have dimensions of each box and the remaining capacity in each box. For example, if a box has length, width & height 5, 7 & 3 respectively, with available capacity of 30 units means the box is filled with 75 units only; whereas total capacity of 105 units (5 x 7 x 3).

Your friend wants to reduce number of boxes by shifting the material inside the other box, if possible. Write if statements and print one of the three statements, written in the sample run. (In the box I have added an example for three boxes)

- l1, w1, h1, l2, w2, h2, l3, w3, h3, c1, c2, c3 (variable are already defined and has values, do not declare or take input)

```
size1 = l1 * w1 * h1;
size2 = l2 * w2 * h2;
size3 = l3 * w3 * h3;
m1 = size1 - c1;
m2 = size2 - c2;
m3 = size3 - c3;
total_m = m1 + m2 + m3;
if (total_m <= size1 || total_m <= size2 || total_m <= size3)
    printf ("Only one box required\n");
else if (m1 < c2 || m1 < c3 || m2 < c1 || m2 < c3 || m3 < c1 || m3 < c2)
    printf ("Two boxes required\n");
else
    printf ("All three boxes required\n");
```

Sample Run:

Only one box required
Two boxes required
All three boxes required

Example:

L1: 7, W1: 7, H1: 3
L2: 9, W2: 8, H2: 5
L3: 9, W3: 3, H3: 6
C1: 1, C2: 32, C3: 112
All three boxes required

- f. Write a program using while loop to print output (given inside the box)

```
c = 'M';
while (c <= 'P'){
    printf ("%c ", c);
    if (i == 1){
        printf ("*\n");
        i = 0;
    }
    else{
        printf ("#\n");
        i = 1;
    }
    c = c + 1;
}
```

M *
N #
O *
P #

Q2. Show complete working of following codes and print output in single line:

- a. a = b = c = 2
while b < 20:

```
print (a, end = ' ')
a = c - b
b = b + 1
c = c + 2
```

(Points: 5)

a	b	c	b < 20
2	2	2	true
0	3	4	true
1	4	6	true
2	5	8	true
...			
17	20	38	false

2 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16

- b. x = 5

(Points: 5)

```

while x > 0:
    print (x, end = ' ')
    t = t - x
    x = x - 2
print (t)

```

5 3 1 -4

t	x	x > 0
5	5	true
0	3	true
-3	1	true
-4	-1	false

c. x = 8, y = 6, z = 5

(Points: 5)

```

if x > y:          #True
    x = x + y      #    x = 8 + 6 = 14
    y = x - y      #    y = 14 - 6 = 8
    x = x - y      #    x = 14 - 8 = 6
if y > z:
    temp = y       #    temp = 8
    z = y          #    z = 8
    y = temp       #    y = 8
print (x, y, z)

```

6 8 8

Q3. Read code and requirement. See output carefully and explain any logical error exist in the code:

Note: You are not required to correct the code. If you try to write the correct code, it has no consideration.

a. Print integers between 1-99 divisible by 3 and 5:

(Points: 5)

3 5 6 9 10 12 15 18 ... 99

```

i = 1
while i < 100:
    if i % 3 == 0:
        print (a, end = ' ')
    if i % 5 == 0:
        print (a, end = ' ')
    i = i + 1

```

15 is divisible by both 3 and 5. Both if statements are independent; therefore, for integer 15, both conditions will become true, and as a result, 15 will be printed twice. Whereas 15 exists only once in the output. The same is the case with multiples of 15, 30, 45, 60, and 75.

b. Print 20 integers between 1-99 divisible by 2 and 3:

(Points: 5)

6 12 18 24 30 36 42 48 54 60

```

i = 1
while i <= 10:
    if i % 2 == 0 and i % 3 == 0:
        print (i, end = ' ')
    i = i + 1

```

The loop is infinite. The if condition is false, and the statement to increment in i is written inside the if block. No change will occur in the variable i, and the loop will continue for an infinite time. The condition $i \% 2 == 0$ is false, because $i \% 2 = 1$, the same is the case with $i \% 3 = 1$

c. Print 20 random alphabets including both capital and small:

(Points: 5)

```

i = 1
while i <= 20:
    c = randint (65, 122)      #122 is ASCII code of z and 65 is ASCII code of A
    print (chr(c), end = ' ')
    i = i + 1

```

The statement `c = randint (65, 122)` will generate random numbers between 65 and 122 (both inclusive). This range includes characters having ASCII codes 91 to 96, which are neither capital nor small alphabets. Therefore, this code will print extra characters (other than alphabets)